

UIDAI Data Hackathon Submission

About

This document presents our analytical submission for the UIDAI Data Hackathon. The objective of this work is to derive meaningful, actionable insights from Aadhaar enrolment and update data using robust statistical and data science techniques. The analysis focuses on understanding temporal and spatial patterns, identifying anomalies, and generating data-driven insights that can support operational monitoring, policy decisions, and service optimization at UIDAI.

The entire analysis is performed using the official Aadhaar enrolment and update datasets provided by UIDAI, ensuring compliance with the hackathon guidelines and data usage policies.

About Team Member :

Name1 : XYZ

Course : MSc Statistics (University of Delhi)

Email :

Name2: ABC

...

...

Problem Statement

UIDAI generates large-scale Aadhaar enrolment and update data across states, districts, and time periods. Due to the scale and operational complexity, the data may exhibit: - Unusual spikes or drops in enrolment/update metrics - District- or state-level deviations from normal patterns - Temporal irregularities caused by operational, technical, or policy-driven changes

The core problem addressed in this project is: > *How can we systematically analyze Aadhaar enrolment and update data to detect anomalies, uncover trends, and generate interpretable insights across time and geography?*

Approach

Our approach combines statistical analysis, time-series techniques, and spatial comparisons to achieve the following: - Establish normal behaviour baselines at state and district levels - Detect anomalies using both temporal and spatial perspectives -

Compare current performance trends on enrollment & updated data- Present insights through clear visualisations suitable for decision-makers

The solution is designed to be: - **Scalable** for large datasets - **Interpretable** for non-technical stakeholders - **Reusable** for ongoing monitoring

Datasets Used

The analysis exclusively uses Aadhaar enrolment and update datasets provided by UIDAI for the hackathon.

Key Data Components

- **Geographical identifiers:** State, District
- **Temporal fields:** Date / Month
- **Operational metrics:** Enrolment count, update count, and related Aadhaar service metrics

Column Usage (Illustrative)

- `state`: State identifier
- `district`: District identifier
- `date`: Daily or monthly timestamp
- `metric`: Aadhaar enrolment or update volume

The data is aggregated at appropriate temporal levels (daily/monthly) depending on the analysis objective.

Methodology

1. Data Cleaning

- Removal of missing and inconsistent records
- Standardisation of state and district names
- Handling of zero and extreme values using logical flags

2. Preprocessing & Feature Engineering

- Aggregation at district–date and state–date levels
- Creation of derived metrics such as:
 - Rolling averages
 - Growth rates
 - Ratios against historical or spatial baselines

3. Temporal Analysis

- Trend analysis to identify seasonality and long-term patterns
- Rolling window statistics to detect sudden changes
- Comparison of current values against historical averages

4. Spatial Analysis

- District-level performance compared against state-level averages
- Identification of districts that significantly deviate from peer behaviour

5. Anomaly Detection Strategy

Multiple complementary rules are applied: - **Temporal anomalies**: Sudden spikes or drops relative to historical behaviour - **Spatial anomalies**: District metrics significantly higher or lower than state averages - **Rule-based thresholds**: Extreme ratios or deviations flagged for review

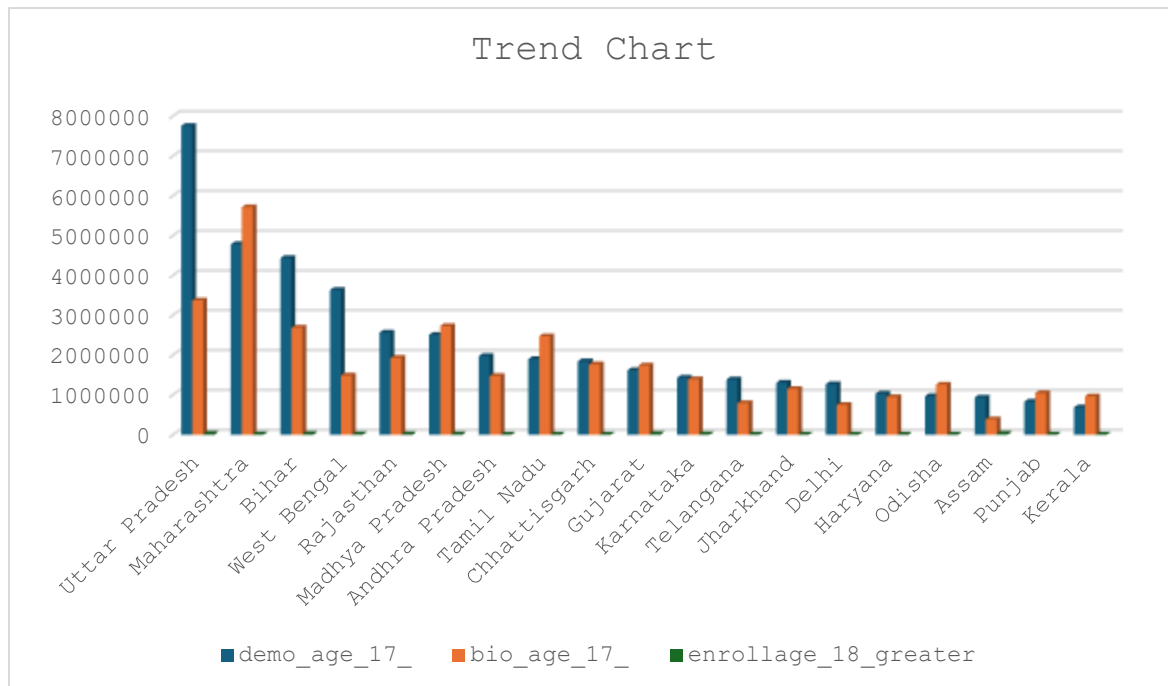
This multi-layered approach reduces false positives and improves robustness.

Data Analysis and Visualisation

Key Insights Generated

- Identification of districts with abnormal enrolment or update activity
- Detection of time periods showing unusual operational behaviour
- Clear differentiation between systematic trends and isolated anomalies

Visualisations



Write insights from the graph :

Code and Notebooks

Link : (optional)